Forced defects in dimer coverings: boundary chemical potential and trapping at frozen edges

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When a planar region cannot be covered completely by dimers, for example a chessboard with two opposite corners removed, uncovered sites (monomers) are forced to exist but are free to move. We ask where they go when all admissible configurations are equally likely, and use their distribution as a parameter-free probe of how a boundary acts on the defects of a constrained system. Exact enumeration on square-lattice domino regions up to 512 sites across shows two regimes: power-law distributions controlled by the corners when the boundary is straight, and exponential trapping in a boundary layer when the boundary is a staircase that freezes the dimers next to it. We introduce a boundary chemical potential, $\mu = \delta \ln Z/\delta h_\partial$, the response of the macroscopic entropy to a shift of the boundary height, which connects the two regimes. On a boundary adjacent to an untilted liquid $\mu = 0$; next to tilted or frozen regions $\mu \neq 0$ and the defect weight acquires a factor $\exp \int \mu \, \mathrm{d}s$. Inside frozen regions μ is not fixed by the limit shape, and we obtain it by continuing the complex Burgers solution along tangent lines to the arctic curve. This reproduces the exact large-deviation law of the Aztec diamond identically, and predicts without free parameters the trapping rate in mixed straight/staircase domains. When the defect path runs along frozen edges, linear response fails; a layer-removal cost $\partial \ln Z/\partial(\text{size})$ with competing nested channels restores agreement. This rule, inferred from the Aztec diamond, predicts exact data for lozenge tilings of hexagons to within 0.5%, including cases where linear response is off by orders of magnitude. In the power-law regime the defects are governed by fractional corner charges and same-sign images, which are treated in a companion paper. We prove the corner-to-boundary law $4/[\pi(x+1)]$, reduce straight-string determinants to a single scalar Fisher–Hartwig symbol, and identify the half-plane and quadrant amplitudes as $2^{3/8}G(\frac{1}{2})^2$ and $2^{9/8}G(\frac{1}{2})^2$.

I. INTRODUCTION

Close-packed dimers on a lattice are among the oldest exactly solvable problems of statistical mechanics [1–3]. They also serve as a paradigm for the physics of local constraints. The constraint that every site is covered exactly once turns the dimer configurations into a divergence-free flux. It endows the model with a height field that is critical and Gaussian at long wavelengths [5–8], and it turns monomers (uncovered sites) into charges of an emergent electrostatics. The same "Coulomb phase" physics appears in three-dimensional dimer models [9], in spin ice, where violations of the ice rule behave as magnetic monopoles [10], and in artificial spin-ice arrays, where such defects can be imaged directly [11, 12]. In all these systems the forces between defects are, at least in part, *entropic*: defects attract or repel only because their positions change the number of available configurations.

The interaction of defects placed at *prescribed* positions is well understood. Monomer pairs in the bulk [3, 4, 13, 31, 32] and on a straight boundary [14] have been computed exactly. Ciucu and co-workers proved that gaps of given charge interact by two-dimensional electrostatics, first for lozenge tilings [15], then on the square lattice [36, 37] and on all doubly periodic bipartite lattices in the liquid phase [38]. Near a boundary the images of a gap have the same sign if the boundary is constrained and the opposite sign if dimers may protrude across it [16, 39, 40]. Monomers pinned at corners correspond to boundary-condition-changing operators [17], and a single vacancy in an odd board has been studied through Temperley's bijection [33–35]. All of this concerns boundaries next to a critical (liquid) dimer phase, where the electrostatic picture applies.

Here we ask a different and very elementary question. Suppose a finite region is unbalanced, so that it *cannot* be tiled completely; the classic example is the "mutilated chessboard" with two opposite corners removed. Defects are then forced to exist, but they are free to move, and nothing except the boundary tells them where to go. If every admissible configuration is equally likely, where are the defects? Their distribution measures directly how a boundary couples to the defects of a constrained system, with no adjustable parameter. The question becomes sharp for boundaries that are *not* next to a critical phase. Staircase boundaries, as in the Aztec diamond, freeze the dimers next to them [18], and there the electrostatic picture gives no prediction at all. We find that the defects are then trapped exponentially in a boundary layer near the points where the arctic curve touches the edge (Sec. V). The main purpose of this paper is to explain this trapping quantitatively and to connect it with the power-law regime of straight boundaries.

Our main results are the following.

(i) *Boundary chemical potential* (Sec. VI). We introduce $\mu(s) = \delta \ln Z/\delta h_\partial(s)$, the response of the macro-

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scopic entropy to a shift of the boundary height. This is precisely the perturbation created by a defect and the string that connects it to the boundary, so to leading exponential order the defect weight is $\exp\int \mu\,\mathrm{d}s$, Eq. (12). On a straight edge next to an untilted liquid, $\mu = 0$ and only the power laws of the Coulomb gas remain; next to tilted or frozen regions, $\mu \neq 0$ and defects are trapped. In frozen regions the limit shape [19, 20] does not determine μ. We fix it by continuing the dual variables of the complex Burgers equation along tangent lines to the arctic curve. For the Aztec diamond this gives $\mu\delta/2 = \ln[(1-\kappa)/\kappa]$, Eq. (15), identical to the exact lattice law, whereas a naive regularization is off by a factor of about 1.45. In a "house"-shaped domain that mixes straight and staircase edges, the same construction predicts the exponential trapping rate and the optimal trapping point without free parameters (Fig. 5).

(ii) *Layer cost and competing channels* (Sec. VI). When the defect path runs along frozen edges, linear response is not enough. Each defect on a frozen staircase forces a zipper of dominoes, and two zippers meeting at a corner remove a layer of the domain. The resulting rule, Eq. (16), adds the layer-removal cost $\partial \ln Z/\partial(\text{size})$ and optimizes over nested channels. Inferred from the Aztec diamond, it predicts exact results for lozenge tilings of hexagons to within 0.5%, including cases where linear response alone is wrong by orders of magnitude (Table I). The layer cost required by consistency coincides with the derivative of $\ln Z$ along the defect-free side of the hexagon, not along the other two sides.

(iii) *Exact results* (Sec. IV). We prove the corner-to-boundary law $4/[\pi(x+1)]$ in the quadrant, Eq. (3). We show that straight-string determinants in the plane and the half plane reduce to Toeplitz and Toeplitz+Hankel determinants of a single scalar symbol, which reproduces the known bulk amplitude and gives the new half-plane and quadrant amplitudes $2^{3/8}G(\frac{1}{2})^2$ and $2^{9/8}G(\frac{1}{2})^2$ to 18 digits.

The power-law regime of straight edges, where corners act as fractional charges and edges as same-sign mirrors, is summarized in Sec. III and treated in detail, together with its dependence on stiffness, lattice and bipartiteness, in the companion paper [43]. Throughout we distinguish rigorous results, results established by exact arithmetic to many digits, numerically supported statements and conjectures; Table II summarizes the status of each result and identifies what was known before.

II. MODEL AND METHODS

A. Dimers, heights and charges

We consider unions of unit squares (cells) of $\mathbb{Z}^2$, colored like a chessboard; cell (x, y) is black if $x + y$ is even. A domino covers one black and one white cell. If the region has n_w white and n_b black cells, at least $|n_w - n_b|$ cells remain uncovered. We study the uniform measure on configurations with exactly the forced number of defects. We call the white defects negative ($e = -1$) and the black defects positive ($e = +1$). A removed cell carries the charge of a defect of its color.

The Thurston height h_T changes by $+1$ (-1) when one walks along a lattice edge with a black (white) cell on the left. Around a covered vertex h_T returns to its value. Around a defect it winds by ± 4, so the defect is a vortex of the height field. We use the normalized height $h = h_T/2$, whose admissible slopes satisfy $|h_x| + |h_y| \leq 1$. In the scaling limit h converges to a Gaussian free field [5, 6], and defects become electric charges of the dual (magnetic) description [7, 8].

B. Exact enumeration

We use three independent exact methods. (a) A row-by-row broken-profile transfer matrix with 128-bit integers, which gives exact counts for all defect configurations on boards up to $N = 16$. It serves as the reference. (b) Kasteleyn determinants [1], with horizontal weights 1 and vertical weights i (or real signs alternating by column). For regions with interior defects, the Kasteleyn condition is restored by flipping the signs of the edges crossed by a "string" joining the defect to the boundary. (c) A low-rank update. If $G = K^{-1}$ is known for the defect-free region, the ratio of partition functions with defects at v and c is the bordered determinant

$$\frac{Z(\Omega \setminus \{c, v\})}{Z(\Omega)} = \left| \det \begin{pmatrix} I - 2G(b_j, w_l) & -2G(b_j, v) \\ G(c, w_l) & G(c, v) \end{pmatrix} \right|, \tag{1}$$

where (b_j, w_l) are the endpoints of the string edges. For the infinite plane, half plane and quadrant, the entries of G are exact numbers of the form $a + b/\pi$ with rational a, b. We generate them by an exact recursion and evaluate determinants in ball arithmetic (Arb) with certified error bounds, up to string lengths $r = 1024$. On 8×8 boards the three methods agree for all 1552 defect configurations. Aztec-diamond ratios were obtained from exact integer inverses (FLINT), because double precision fails beyond $n \simeq 44$ (the condition number reaches $\sim 10^{17}$ at $n = 64$). All key claims of Secs. IV–VI were independently re-derived from scratch by a separate code base; no discrepancy was found.

III. STRAIGHT EDGES: THE POWER-LAW REGIME

On a straight edge next to an untilted liquid, $\mu = 0$ (Sec. VI) and the defects feel only power laws. Walking counterclockwise along the boundary, the height oscillates between two values on a straight edge, so its average is constant along each edge and jumps at corners: by $J_c = +1$ at a black corner, -1 at a white corner and $-3 \equiv +1$ (modulo one monomer's winding) at the

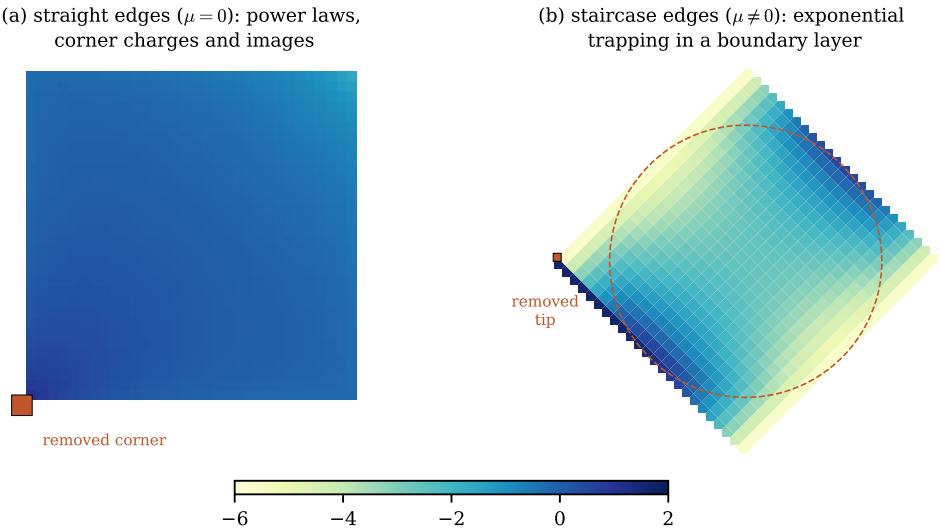


FIG. 1. Two regimes for a forced defect (exact densities, common logarithmic scale). (a) A 64×64 board with one corner removed: the boundary is straight and next to an untilted liquid, $\mu = 0$, and the density varies only by power laws set by the corners [43]. (b) The Aztec diamond of order $n = 24$ with its west tip removed: the staircase boundary freezes the dimers next to it (dashed: arctic circle), $\mu \neq 0$, and the defect is trapped in a boundary layer; the density varies over eight decades.

notch of a removed black corner. In the companion paper [43] we show that these jumps act on mobile defects as fixed fractional charges $q_c = -J_c/4$ (cf. the boundary-condition-changing operators of Allegra [17]), and that each edge is a mirror with images of the same sign [39]. The resulting conformally covariant density,

$$\rho(z) \propto |f'(z)|^{1/4}\,[\mathrm{Im} f(z)]^{1/4} \prod_c |f(z) - f_c|^{e\,q_c}, \qquad (2)$$

with f a conformal map onto the upper half plane, reproduces exact densities on rectangles to 1–3% [Fig. 1(a)], including the $r^{-3/4}$ divergence next to a removed corner and the $N^{-1/4}$ depletion of the corners of odd boards. Pair correlations decay as d^{-1} along a boundary and as $d^{-1/2}$ in the bulk; their amplitudes are derived in Sec. IV. Here these power laws serve as the baseline against which the exponential trapping of frozen boundaries is measured.

IV. EXACT RESULTS AND AMPLITUDES

A. Corner to boundary

Theorem 1 (Quadrant, boundary defect)**.** *In the infinite quadrant with its corner cell c removed, place a white defect on the boundary cell $(x, 0)$, with x odd. Then, for $x \geq 3$,*

$$\begin{aligned} \frac{Z(Q \setminus \{c, (x,0)\})}{Z(Q)} &= \frac{4}{\pi} \int_0^{\pi/2} \sin\theta\, \sin[(x+1)\theta] \\ &\qquad \times \sqrt{1 + \cos^2\theta}\, \mathrm{d}\theta \\ &= \frac{4}{\pi(x+1)} + O(x^{-3}). \end{aligned} \qquad (3)$$

The values are of the form "integer minus rational/π", e.g. $2 - 16/(3\pi)$ at $x = 3$.

The proof diagonalizes the box Kasteleyn matrix by sine functions and evaluates one integral in closed form. The leading term comes from $\theta = \pi/2$, and the next order vanishes by parity. The exponent -1 is the Coulomb-gas value $|f'|^{1/2}|f|^{-3/4}$ for $f = z^2$, i.e. a boundary defect next to the removed-corner charge $q_c = +\frac{3}{4}$ [43], while the prefactor $4/\pi$ is a lattice quantity.

B. A single scalar symbol

For a vertical string, the horizontal offset between the black end of the jth string edge and the white end of the lth depends only on the parity of $j - l$. Hence $I - 2G$ is a Toeplitz matrix in $j - l$. Summing the Fourier series in closed form gives the symbol

$$\sigma(e^{i\varphi}) = \mathrm{sgn}(\cos\varphi)\, \frac{\cos\varphi + i}{\sqrt{1 + \cos^2\varphi}}, \qquad (4)$$

which is a pure phase with jumps of π at the massless points $\varphi = \pm\pi/2$ (Fig. 2b). Moreover, when the defect sits at the end of the string and the removed cell at its

start, the border row of Eq. (1) equals $-\frac{1}{2}$ times the first row. It can therefore be eliminated exactly. With c_m the Fourier coefficients of σ,

$$R_{\mathrm{bulk}}(r) = \tfrac{1}{2}\left|\det\left[c_{j-k-1}\right]_{j,k=0}^{r-1}\right|, \tag{5}$$

$$R_{\mathrm{half}}(r) = \tfrac{1}{2}\left|\det\left[c_{j-k-1} - c_{j+k+3}\right]_{j,k=0}^{r-1}\right|, \tag{6}$$

for a pair at vertical distance r in the plane, and for a boundary cell and a defect at height r above it in the half plane. The Hankel part in the second line is the lattice image charge.

C. Amplitudes

The symbol $z\sigma$ is of Fisher–Hartwig type with $\beta_1 = \beta_2 = \frac{1}{2}$ at $z = \pm i$. The Deift–Its–Krasovsky asymptotics [21] then gives

$$R_{\mathrm{bulk}}(r) \simeq \pi\, 2^{-3/4}\, G(\tfrac{1}{2})^4\, r^{-1/2} = 0.2473718\ldots\, r^{-1/2}. \tag{7}$$

The factors are $2^{-1/4}$ from $\sum_k kV_kV_{-k} = -\operatorname{artanh}(q^2)$ with $q = \sqrt{2}-1$, $2^{1/2}$ from $|z_1-z_2|^{2\beta_1\beta_2}$, and $[G(\frac{3}{2})G(\frac{1}{2})]^2$ from the two jumps. This axis amplitude was first obtained in closed form by Wilkins and Powell [32] from a Toeplitz representation derived with the transfer matrix; their $E/2 = 2^{-7/12}e^{6\zeta'(-1)}$ equals $\pi 2^{-3/4}G(\frac{1}{2})^4$. The same constant was obtained for diagonal separation by Hartwig [4] ($B/4$ with $B = 0.98948729$). The scalar symbol reproduces both and extends directly to the half plane and the quadrant. For the half plane and the quadrant we established the amplitudes by certified arithmetic up to $r = 1024$ and extrapolation, to 18–19 digits (Fig. 2c):

$$C_{\mathrm{half}} = 2^{3/8}G(\tfrac{1}{2})^2, \qquad C_{\mathrm{quad}} = 2^{9/8}G(\tfrac{1}{2})^2. \tag{8}$$

The pattern "lattice value = operator normalization × continuum Coulomb-gas correlator" accounts for all three. The transcendental factor $G(\frac{1}{2})^2$ belongs to each bulk monomer, while the boundary and corner normalizations involve only π and powers of 2. A proof for the half plane would require the asymptotics of a Toeplitz+Hankel determinant with a *non-even* Fisher–Hartwig symbol $z^{-1}\sigma$ of winding -1. This lies just outside the existing theory [21–23].

V. STAIRCASE BOUNDARIES: TRAPPING IN THE AZTEC DIAMOND

The Aztec diamond AD_n has $Z = 2^{n(n+1)/2}$ tilings [24]. Four frozen corners are separated from a liquid disk by the arctic circle [18]. We remove the west tip cell t, which forces one white defect, and we also consider a missing domino, which forces two defects of opposite colours. In both cases the defects avoid the liquid region. Instead they stay within a boundary layer about two cells thick, near the points where the arctic circle touches the edges (Fig. 3a). The weight outside this layer decays as n^{-1} for one defect and as $n^{-1/2}$ for a pair (Fig. 3c).

On the boundary the distributions are known exactly. Let k be the position of the defect along the edge, counted from the tip. For the adjacent (SW) and the opposite (NE) edge,

$$\frac{Z(\mathrm{AD}_n \setminus \{t, \mathrm{SW}_k\})}{Z(\mathrm{AD}_n)} = P\big(\mathrm{Bin}(n, \tfrac{1}{2}) \geq k\big), \tag{9}$$

$$\frac{Z(\mathrm{AD}_n \setminus \{t, \mathrm{NE}_k\})}{Z(\mathrm{AD}_n)} = \binom{n-1}{k-1} 2^{-n}, \tag{10}$$

For a missing domino with defects at positions k and l on the two edges adjacent to a corner,

$$\frac{Z(\mathrm{AD}_n \setminus \{\mathrm{S1}_k, \mathrm{S2}_l\})}{Z(\mathrm{AD}_n)} = \sum_{j=0}^{n-1} 2^{j-1}\, P\big(\mathrm{Bin}(j, \tfrac{1}{2}) = k-1\big) \times P\big(\mathrm{Bin}(j, \tfrac{1}{2}) = l-1\big). \tag{11}$$

Equation (11) is equivalent to the boundary generating function of Chhita and Young [25], and Eqs. (9)–(10) are special cases of it. We found these laws independently by exact computation. We also give elementary proofs of Eqs. (9) and (10) that do not use urban renewal (Appendix A). Equation (11) has a transparent dynamical reading. In domino shuffling the defect pair is created at the corner at "time" j with weight $\frac{1}{2}$, and afterwards the two defects perform independent random walks along their edges. Summing the pair weights over all boundary positions gives $2^{n+1} - 2 = Z(\mathrm{AD}_{n+1})/Z(\mathrm{AD}_n) - 2$.

VI. BOUNDARY CHEMICAL POTENTIAL

A. Definition

A defect and the removed cell are joined, along the boundary, by an arc Γ on which the boundary height is shifted by $\delta = 2$ (one monomer of Thurston winding). To leading exponential order the defect weight is the change of the macroscopic entropy. For a small shift this is the linear response

$$\ln w \simeq \delta \int_\Gamma \mu(s)\, \mathrm{d}s, \qquad \mu(s) = \frac{\delta \ln Z}{\delta h_\partial(s)} = \mathbf{n}\cdot\nabla_p\, \mathrm{ent}(\nabla h)\big|_\partial, \tag{12}$$

where ent is the Cohn–Kenyon–Propp local entropy [19], h is the limit shape and $\mathbf{n}$ is the outward normal. On a flat, liquid-adjacent edge such as that of the chessboard, $\nabla h = 0$ gives $\mu = 0$. The exponential factor is then absent and only the Coulomb-gas power laws of Sec. III remain. On staircase edges, and on straight edges adjacent to a *tilted* liquid, $\mu \neq 0$ and the defects are trapped exponentially.

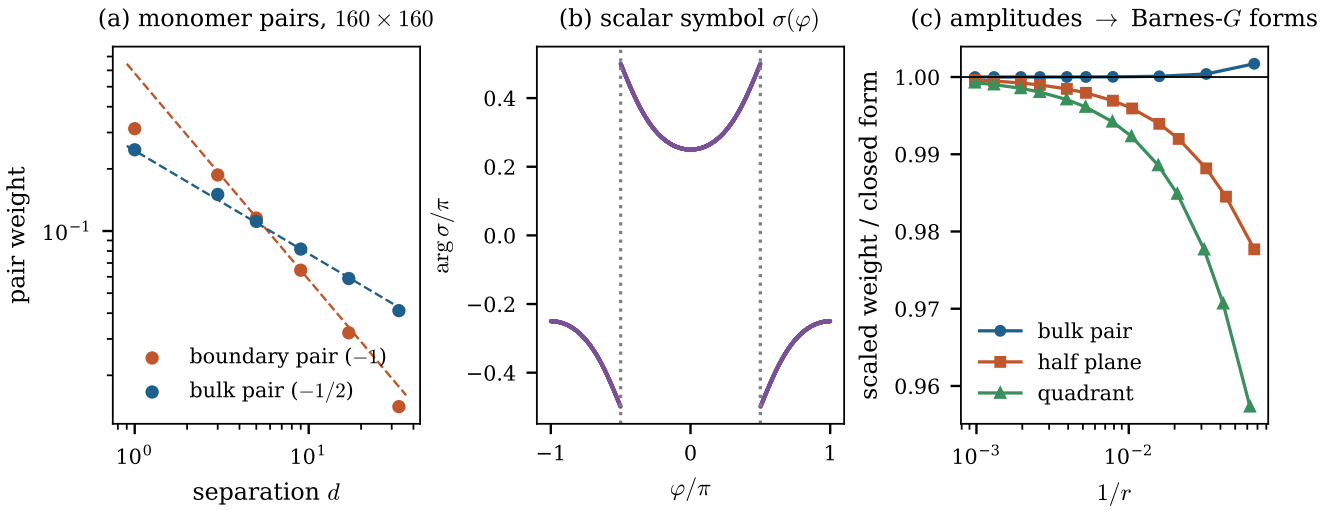


FIG. 2. (a) Monomer-pair correlations on a 160 × 160 board: along the boundary ($\propto d^{-1}$, same-sign images) and in the bulk ($\propto d^{-1/2}$). (b) Phase of the scalar symbol σ of Eq. (4), with jumps of π at $\varphi = \pm\pi/2$. (c) Exact weights, scaled by the expected power law and divided by the closed forms $\pi 2^{-3/4} G(\frac{1}{2})^4$ (bulk), $2^{3/8} G(\frac{1}{2})^2$ (half plane) and $2^{9/8} G(\frac{1}{2})^2$ (quadrant), versus $1/r$.

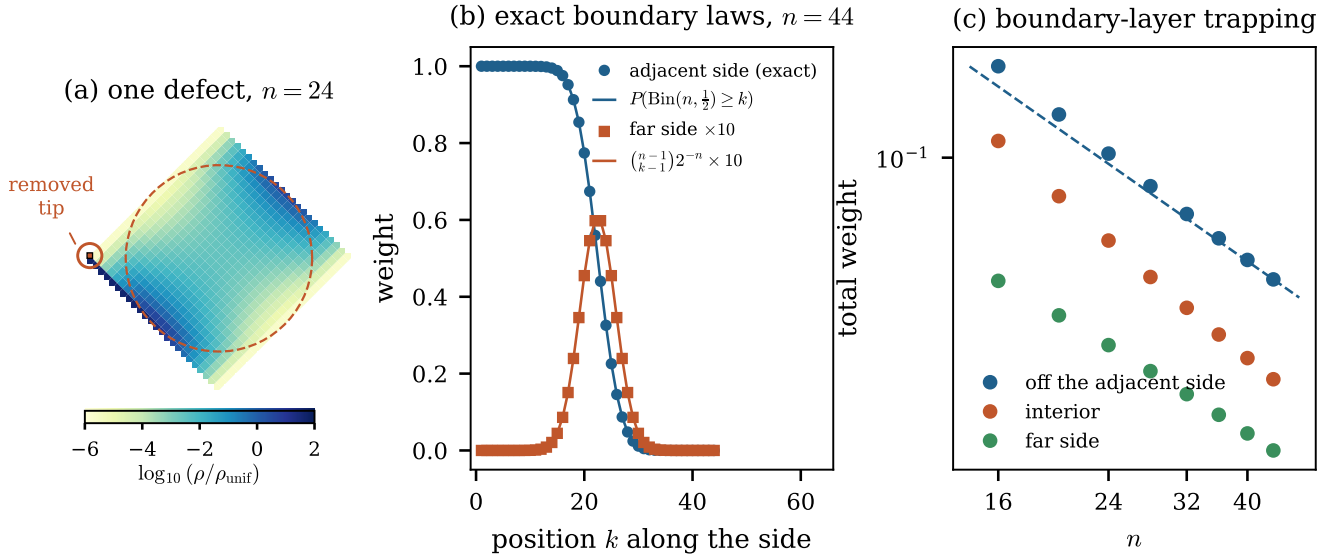


FIG. 3. Aztec diamond with the west tip removed. (a) Exact defect density $\rho/\rho_{\mathrm{unif}}$ for $n = 24$ on a logarithmic colour scale spanning eight decades; each cell of the defect sublattice is drawn as a diamond reaching its four neighbours. The dashed line is the arctic circle, and the small circle marks the removed tip. The density falls to 10^{-6} along the two edges that do not touch the removed tip and drops steeply near the far end of the adjacent edge, following the binomial tail of Eq. (9). (b) Exact weights along the adjacent (south-west) and opposite edges for $n = 44$ (symbols), compared with Eqs. (9)–(10) (lines). (c) Total weight off the adjacent edge, in the interior and on the far edge, versus n.

B. Frozen boundaries: tangent continuation

In a frozen region the minimizer h lies on the boundary of the allowed slope polygon, so it does not determine $\nabla_p \mathrm{ent}$. Consistently, a penalty-regularized finite-element solution gives values of μ that depend on the regularization, by factors of order 1.5 (Fig. 4c). The missing information is supplied by the complex Burgers structure of dimer limit shapes [20]. The dual variables $(\ln|z|, \ln|w|)$ are constant along straight characteristics. In frozen regions these characteristics are the tangent lines to the arctic curve.

For the uniform Aztec diamond ($|x|+|y| \leq 1$) the local

dimer probabilities are known in closed form [26]:

$$p_{E,W} = \frac{1}{2} + \frac{1}{\pi}\arctan\frac{\pm 2x-1}{\sqrt{1-2x^2-2y^2}}, \qquad (13)$$

and similarly $p_{N,S}$ with y. Their ratios have finite limits on the arctic circle. The tangent from the edge point $P(\kappa) = (-1+\kappa, -\kappa)$ touches the circle at

$$Q(\kappa) = \frac{\left(2\kappa^2-1,\ 2\kappa^2-4\kappa+1\right)}{2(2\kappa^2-2\kappa+1)}. \qquad (14)$$

Projecting the dual variables at Q onto the edge normal gives

$$\frac{\delta}{2}\,\mu(\kappa) = \frac{1}{2}\ln\left[\left(\frac{1-\kappa}{\kappa}\right)^2\right] = \ln\frac{1-\kappa}{\kappa} \qquad (15)$$

identically in κ. This is exactly the lattice value $\ln[(n-k)/k]$ that follows from Eqs. (9)–(11). The limit shape plus tangent continuation thus reproduces the full large-deviation profile $n[H(\kappa) - \ln 2]$ of the Aztec diamond (Fig. 4). The same procedure applied to the arctic curve extracted from a finite-element solution reproduces Eq. (15) to 1–3%.

C. A mixed domain

As a test with no free parameter we consider a "house": an $N\times N$ square whose two upper corners are cut by staircases of opposite colours (size k). The staircases balance the colours, and removing a1 creates one defect. Exact integer computations up to $N = 128$ show that the defect leaves a1, the Coulomb attractor, and is trapped on the staircase of the same colour with weight $w \simeq e^{RN}N^{-1}$. The exact rates are $R = 0.213$ for $k = N/2$ and $R = 0.255$ for $k = N/4$. We computed the limit shape by finite elements. We evaluated μ directly where the boundary is liquid-adjacent and by tangent continuation where it is frozen, and integrated Eq. (12) along Γ. This predicts $R = 0.204$–0.217 and 0.259–0.271 (range over meshes $N = 48$–96), with the optimum at $x/N = 0.07$–0.09 (exact: 0.08–0.09), as shown in Fig. 5. Most of the gain comes not from the staircase but from the straight left edge, because the staircases tilt the adjacent liquid. The optimum sits where μ changes sign on the staircase.

D. Beyond linear response: layer cost and competing channels

Confronting Eq. (12) with the exact Aztec laws Eqs. (9)–(11) reveals two further ingredients. They matter whenever Γ runs along *frozen* edges. First, the local gradient of $\ln w$ is always given correctly by Eq. (15), but the rate itself is smaller than $\int_\Gamma \mu$ by exactly $\ln 2$. Second, for two defects close to a corner the dominant contribution comes from $j < n$ in Eq. (11). All three families are described by the single rule

$$\frac{1}{n}\ln w = \max_{\tau}\Big[\tau\, L\Big(\frac{\kappa}{\tau},\frac{\lambda}{\tau}\Big) - \frac{1}{n}\,\frac{\partial \ln Z(\Omega_m)}{\partial m}\Big|_{m=\tau n}\Big], \qquad (16)$$

where $\Omega_{\tau n}$ is the scaled copy of the domain sharing the relevant corner, L is the linear response $\int_\Gamma \mu$ computed in the unit domain, and the second term is the entropy cost of removing one lattice layer. For the Aztec diamond the layer cost is $\tau \ln 2$. Only $\tau = 1$ is allowed if an endpoint lies on an edge not containing the corner. Equation (16) reproduces Eqs. (9) (optimum $\tau = \min(2\kappa, 1)$), (10) and (11) exactly. At the lattice level, a defect on a frozen staircase forces a zipper of dominoes (Appendix A). Two zippers meeting at the corner strip one layer off the inner domain, and the turning point of the zippers is τ. When Γ lies entirely on liquid-adjacent edges no zipper forms and Eq. (16) reduces to linear response, which is the situation of the house domain above.

E. Independent test: lozenge tilings of hexagons

To test Eq. (16) away from the data that suggested it, we studied lozenge tilings of a hexagon with side lengths a (bottom), b (right) and c. We remove an up-triangle at distance k and a down-triangle at distance l from the bottom-right corner. By the Lindström–Gessel–Viennot lemma [27, 28], the number of tilings is a $(c+1)\times(c+1)$ determinant of binomial coefficients. The sources are $(-i, i)$ and $(a-k, 1)$, and the sinks are $(a-j, b+j)$ and $(a-1, l)$. We verified this formula against Kasteleyn determinants for all dent positions in three hexagons, and it allows exact computation up to $n = 240$. Dent-pair counts on the boundary of a hexagon are also given in closed form by Ciucu and Fischer [41]; the limit shape is the arctic circle of Cohn, Larsen and Propp [42].

With the corner triangle fixed, the dent position follows $P(X = m) \propto \binom{a-1+m}{m}\binom{c-1+b-m}{b-m}$. Consistency of Eq. (16) then *requires* a layer cost equal to $\max f - f(0)$, where f is the large-deviation function of X. For five hexagons this number equals $\partial \ln Z/\partial c$ to three or four digits and differs clearly from $\partial_a \ln Z$ and $\partial_b \ln Z$ (Fig. 6b). The removed layer is therefore the one in the defect-free direction. With no further input, Eq. (16) then predicts the pair rates, including channels with $\tau \approx 0.24$–0.76 where linear response alone fails completely (e.g. 0.003 instead of 0.139). The agreement with the exact values is within 0.1–0.5% for nine cases in three hexagons (Table I). Along the diagonal $\kappa = \lambda$ it holds at twenty further points in two hexagons, with deviations below 0.3% (Fig. 6a).

VII. DISCUSSION AND CONCLUSIONS

Forced defects in a dimer covering are moved only by entropy, and the boundary is the only thing that

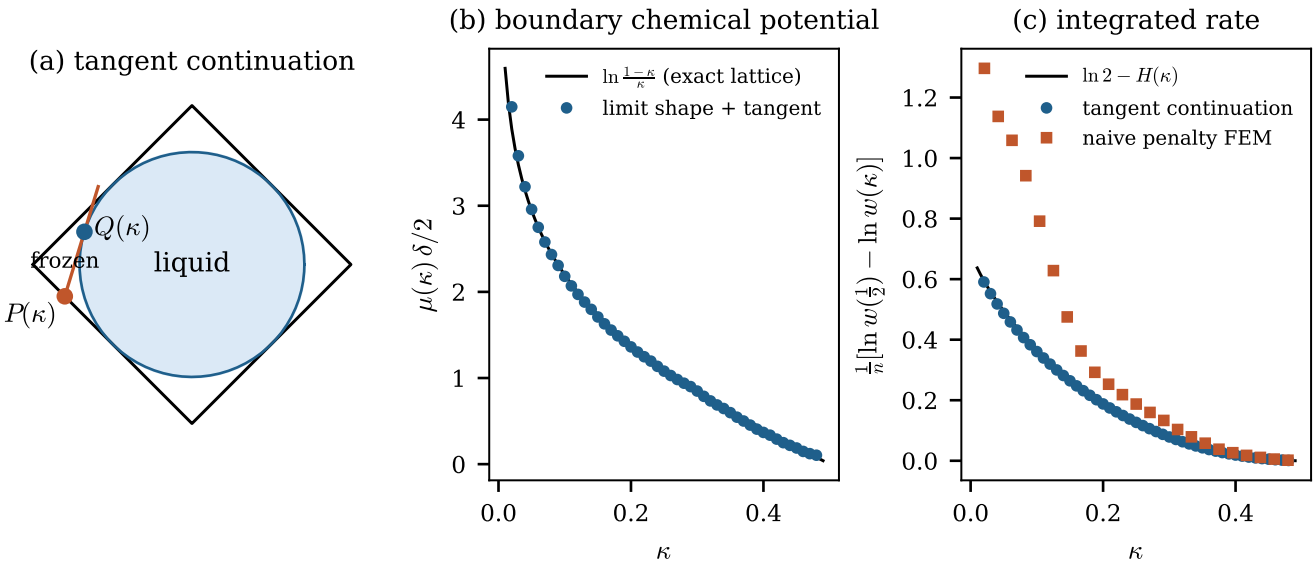


FIG. 4. Boundary chemical potential of the Aztec diamond. (a) The dual variables are constant along tangent lines to the arctic circle, so μ at the edge point $P(\kappa)$ is set by the tangency point $Q(\kappa)$. (b) $\mu(\kappa)$ from the limit shape with tangent continuation (symbols, extracted from a finite-element limit shape) against the exact lattice value $\ln[(1-\kappa)/\kappa]$ (line). (c) Integrated rate: exact continuum result (line), tangent continuation (circles) and the naive penalty-regularized linear response (squares), which is biased by a factor of about 1.45.

TABLE I. Pair rates $\frac{1}{n}\ln w$ for dents at distances (κ,λ) from a corner of the hexagon $H(a{:}b{:}c)$. Columns: linear response only ($\tau=1$), the rule Eq. (16) with its optimal τ, and exact values extrapolated from $n=40$–240 with $a+b\ln n/n+c/n$.

$a{:}b{:}c$	(κ,λ)	$\tau=1$	rule (τ^*)	exact
1:1:1	$(0.1, 0.1)$	0.003	0.1387 (0.37)	0.1386
1:1:1	$(0.2, 0.2)$	0.259	0.2774 (0.75)	0.2766
1:1:1	$(0.1, 0.3)$	0.208	0.2251 (0.76)	0.2244
1:1:1	$(0.3, 0.3)$	0.412	0.4124 (1)	0.4111
1:1.5:0.75	$(0.1, 0.1)$	−0.201	0.1387 (0.24)	0.1382
1:1.5:0.75	$(0.15, 0.3)$	0.172	0.2866 (0.52)	0.2865
1:1.5:0.75	$(0.3, 0.6)$	0.573	0.5728 (1)	0.5718
1.4:0.8:1	$(0.3, 0.15)$	0.268	0.2866 (0.75)	0.2850
1.4:0.8:1	$(0.6, 0.4)$	0.554	0.5544 (1)	0.5534

tells them where to go. Their distribution is therefore a clean, parameter-free measure of how a boundary acts on the defects of a constrained system. The central result of this paper is that this action is captured by a single local quantity, the boundary chemical potential $\mu = \delta \ln Z/\delta h_\partial$, the entropy cost of shifting the boundary height. Next to an untilted liquid μ vanishes; next to tilted or frozen regions it does not, and the defects are trapped exponentially. In frozen regions μ is fixed by continuing the complex Burgers solution along tangent lines to the arctic curve. With this one ingredient the Aztec-diamond law is reproduced identically, and the trapping rate and optimal trapping point in mixed domains are predicted without parameters. Along frozen edges a geometric layer cost and a competition between nested channels must be added. The resulting rule, Eq. (16), was inferred from the Aztec diamond and predicts exact lozenge-tiling data to within half a per cent, in cases where linear response fails by orders of magnitude.

In the opposite limit, $\mu = 0$, only the power laws of the emergent electrostatics remain, and the boundary acts through its corners as fractional charges and through its edges as same-sign mirrors [43]. At the lattice level, the amplitudes factorize into a universal bulk monomer normalization, which contains $G(\frac{1}{2})^2$, and elementary boundary and corner normalizations, and straight-string determinants reduce to a single scalar Fisher–Hartwig symbol.

The same questions arise wherever constrained degrees of freedom produce emergent Coulomb physics with boundaries. Examples are monopoles in finite spin-ice samples and nanostructures, defects in artificial spin ice, where edge and corner populations can be imaged by magnetic force microscopy [11, 12], dimer adsorbates and molecular monolayers on terraces, and constrained colloidal systems. Our results suggest that defects should be trapped exponentially wherever the boundary orders (freezes) the constrained phase next to it, at the points where the ordered and critical regions meet, with a rate set by the boundary chemical potential. Where the phase touching the boundary is critical, corners should instead act as fractional charges and constrained edges as same-sign mirrors. How the corner charges depend on the

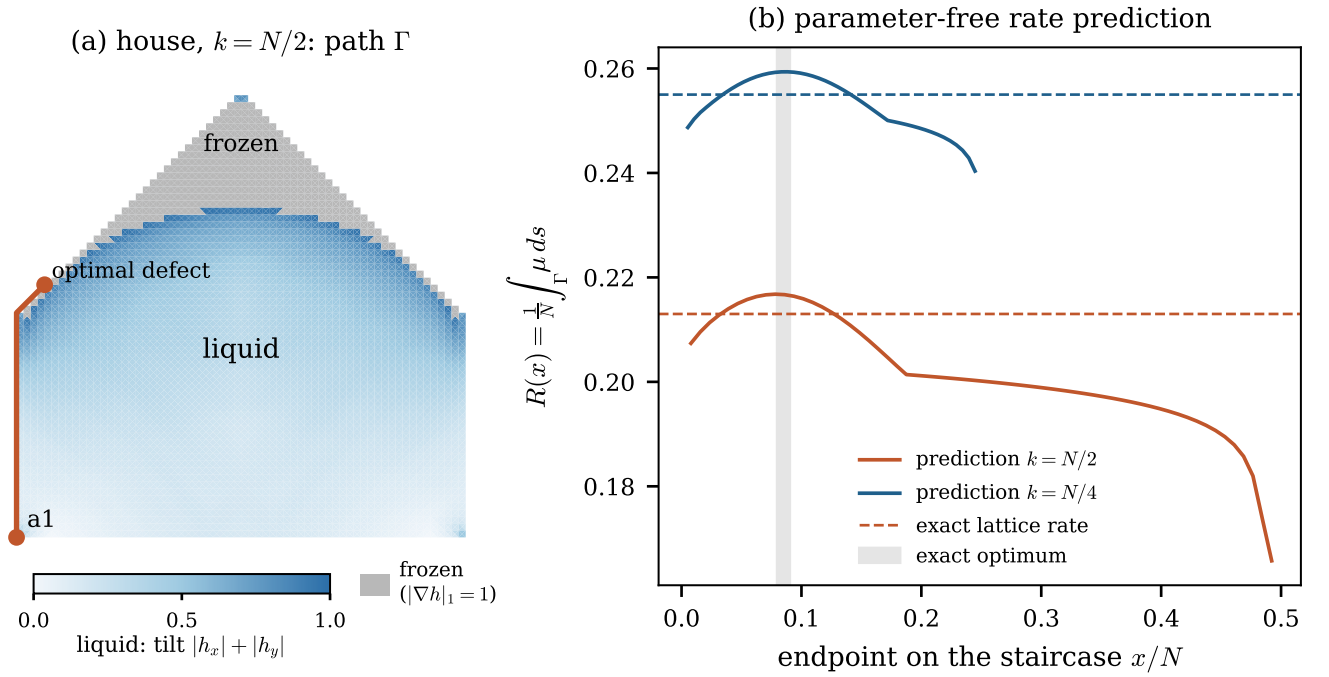


FIG. 5. House-shaped domain. (a) Limit shape ($k = N/2$) from the finite-element solution. Grey marks frozen regions, where the slope lies on the boundary of the allowed polygon, $|h_x| + |h_y| = 1$; in the liquid the shade gives the tilt $|h_x| + |h_y|$, which controls μ on the adjacent boundary. Also shown is the defect path Γ from the removed corner a1 along the straight edge to the optimal point on the staircase. (b) Predicted rate $R(x) = N^{-1}\int_\Gamma \mu\, ds$ versus the endpoint x/N for $k = N/2$ and $N/4$ (solid), compared with the exact lattice rates (dashed) and the exact optimal position (grey band).

stiffness, the lattice and bipartiteness is the subject of Ref. [43].

Several questions remain open. Among them are the Toeplitz+Hankel asymptotics that would prove the half-plane and quadrant amplitudes. Others are a first-principles justification of the tangent continuation and of the layer rule in general domains, which appears to be related to the tangent method of Colomo and Sportiello [30] (proved for a class of six-vertex domains by Aggarwal [44]), and the behaviour of many defects, where screening of the corner charges is expected.

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Appendix A: Elementary proofs for the Aztec diamond

Boundary switch. Each white cell SW_k on the south-west edge has only two neighbours, b_{k-1} and b_k, on the adjacent diagonal. If SW_k is matched to b_k, then SW_{k+1} must also be matched to b_{k+1}. The edge therefore consists of m dominoes of one orientation followed by $n - m$ of the other, and the switch position m is unique.

Sliding bijection. If the tip b_0 is removed, the cells

TABLE II. Summary of results and their status. "Proof": rigorous; "exact": exact arithmetic or ≥ 10 digits; "numerical": agreement at the per-cent level across sizes; "known": previously established in the cited work.

Result	Status
Quadrant boundary law (3)	proof
Scalar-symbol reductions (6)	proof
Bulk amplitude $\pi 2^{-3/4} G(\frac{1}{2})^4$	known: axis [32], diagonal [4]; new derivation
Half-plane, quadrant amplitudes	exact (18 digits)
Aztec laws (9)–(11)	known [25]; new elementary proofs of (9), (10)
μ of the Aztec diamond, Eq. (15)	exact identity (given tangent continuation)
House trapping rates	numerical (5%)
Layer rule (16)	conjecture; exact for Aztec, 0.5% for lozenges

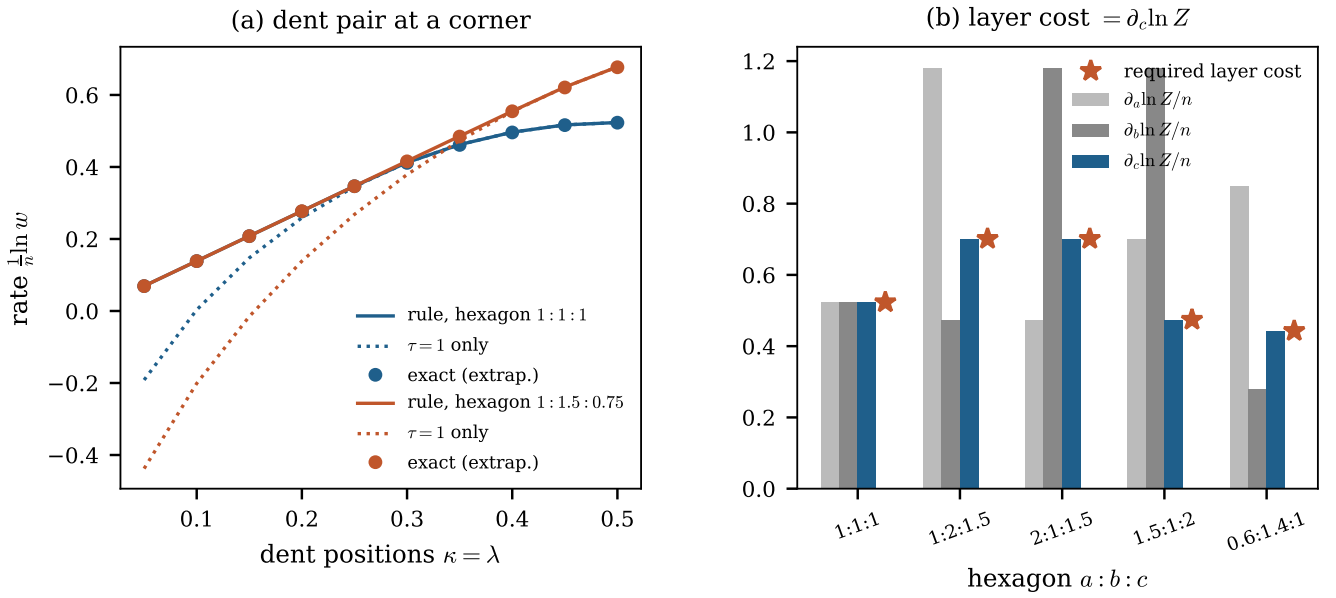


FIG. 6. Independent test on lozenge tilings. (a) Rate of a dent pair at distances $\kappa = \lambda$ from a corner, for two hexagons: exact values extrapolated from $n = 60, 120, 240$ (symbols), the rule Eq. (16) (solid) and linear response alone, $\tau = 1$ (dotted). (b) The layer cost required for consistency (stars) coincides with $\partial_c \ln Z/n$, the derivative with respect to the side length in the defect-free direction, and not with ∂_a or ∂_b.

$\mathrm{SW}_1, \ldots, \mathrm{SW}_{k-1}$ are forced. Sliding the dominoes SW_i–b_{i-1} to SW_i–b_i for $i < k$ gives a bijection between tilings of $\mathrm{AD}_n \setminus \{t, \mathrm{SW}_k\}$ and tilings of AD_n with switch $m \geq k$.

Switch distribution. Let I be AD_n minus the two opposite boundary double layers. Removing a corner cell c of I forces a zipper of $n-2$ dominoes, and what remains is a translate of AD_{n-1} minus its boundary layer. Kuo's graphical condensation [29], applied to $G = I \cup \{b_m, e_t\}$ with four boundary vertices of alternating colour, together with the switch of the opposite layer, gives $T_n(m) = 2^{n-1}[T_{n-1}(m) + T_{n-1}(m-1)]$. Hence $T_n(m) = \binom{n}{m} 2^{n(n-1)/2}$, which proves Eq. (9).

Opposite edge. By 180° rotation, removing the east tip forces the whole north-east double layer. The south-west switch then leaves exactly two free cells $b_s, b_{s'}$ with $s < j \leq s'$. Kuo's identity with two same-colour pairs gives $Z(I \cup \{b_s, b_{s'}\}) = 2^{(n-1)(n-2)/2} \binom{n}{s} \binom{n}{s'} \frac{s'-s}{n}$, and the binomial identity $\sum_{s<j\leq s'} \binom{n}{s}\binom{n}{s'}(s'-s) = n 2^{n-1} \binom{n-1}{j-1}$ yields Eq. (10).

Appendix B: Numerical details

Limit shapes were computed by minimizing the Cohn–Kenyon–Propp functional with the domino entropy written in terms of the Lobachevsky function. We used a finite-element mesh with four triangles per cell, which avoids locking at the slope constraint, and penalty continuation. On the Aztec diamond this reproduces the entropy to 0.02% and the frozen area fraction $1 - \pi/4$. Rates were extrapolated using $a + b \ln n/n + c/n$.

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